Lakshya Jain

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Professional Summary

AI/ML Engineer and Computer Science student skilled in **Python, machine learning, deep learning, NLP, LLMs, and RAG**. Experienced in building and evaluating **end-to-end AI systems** across computer vision, retrieval-augmented generation, recommendation systems, and multi-agent reinforcement learning. Strong hands-on experience in **feature engineering, model optimization, experimentation, cross-validation, and performance evaluation**, with a focus on developing reliable and scalable AI solutions.

Technical Skills

Machine Learning: Machine Learning, Predictive Modeling, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Classification, Clustering, Scikit-learn, XGBoost, LightGBM
Deep Learning & AI: PyTorch, Neural Networks, Computer Vision, NLP, LLMs, Transformers, RAG, Explainable AI, TF-IDF, Attention Mechanisms, Reinforcement Learning, Multi-Agent Reinforcement Learning, PPO, RLlib
AI Engineering: LangChain, LangGraph, Prompt Engineering, Generative AI, Agentic AI, Hugging Face, LLM Evaluation, Semantic Search, Vector Search, BM25, Reranking, MLOps, FAST API, MCP, n8n, Agents, LlamaIndex, LangSmith
Programming & Data: Python, SQL, Pandas, NumPy, Statistical Analysis, EDA, Time-Series Analysis, MySQL, PostgreSQL
Engineering & Cloud: Git, Docker, AWS, Oracle Cloud Infrastructure (OCI), ETL Pipelines, Data Validation, Data Modeling

Professional Experience

Research Intern May 2026 – Present
Deep Learning & Computer Vision Research

- Achieved **83% classification accuracy** and **79% macro F1-score** by designing and implementing a multi-branch **PyTorch deep learning architecture** integrating spatial, contextual, interaction-aware, and wavelet-based feature learning on PlantDoc.
- Achieved up to **99% accuracy** and **98% F1-score** on specific disease categories by developing custom **attention-based feature extraction, cross-attention fusion, and spectral encoding** modules for fine-grained recognition.
- Improved classification robustness through **species-aware hierarchical classification, multi-scale feature learning, and model optimization**, reducing confusion between visually similar plant diseases.
- Validated model performance using **cross-validation and ablation studies** and applied **Grad-CAM, LIME, and SHAP** to improve model interpretability and analyze prediction behavior.

Microsoft Elevate & AICTE Feb 2026 – Mar 2026
Power BI Data Analytics Intern

- Achieved **28% faster reporting** by developing automated **Power BI dashboards** and KPI workflows for SLA monitoring and operational analytics.
- Achieved **99% reporting accuracy** across **15,000+ records** by implementing **SQL validation workflows** and star-schema data models, improving data quality and reliability.
- Optimized **SQL queries** and automated reporting workflows while translating business requirements into analytical dashboards and actionable insights.

Technical Projects

Adaptive Corrective Self-RAG (ACSRAG) — LLM & Retrieval Engineering

- Improved **answer correctness by 32%** by engineering a corrective RAG pipeline with adaptive query rewriting and web-search fallbacks for ambiguous and complex queries.
- Improved **retrieval relevance by 41%** by combining **vector search and BM25** using Reciprocal Rank Fusion (RRF) and Cross-Encoder Reranking to prioritize relevant evidence.
- Reduced **context consumption by 65%** through dynamic context compression that extracted highly relevant passages, improving LLM efficiency and reducing contextual noise.
- Reduced **hallucination rate by 54%** through claim-evidence verification and iterative re-retrieval, validating generated responses against retrieved context.

Multi-Objective Traffic Signal Optimization — Multi-Agent Reinforcement Learning

- Achieved **93.1% reduction in waiting time**, **22.5% reduction in CO_2 emissions**, and **22.5% reduction in fuel consumption** versus fixed-time control using **PPO-based MARL** across a 4-intersection SUMO network.
- Designed forecast-aware agent states combining **per-lane queue lengths, waiting times, signal phases**, and **5-step ARIMA(5,1,0) traffic forecasts** for proactive signal-control decisions.
- Implemented a configurable **multi-objective reward function** balancing waiting time, CO_2 emissions, and fuel consumption through weighted normalization and reward shaping.
- Built an end-to-end **RLlib, SUMO, and TraCI** pipeline with shared-policy training, fixed-time baselines, checkpointing, experiment tracking, and multi-episode evaluation.

Virtual Experience Programs

BCG – Data Science Advanced Simulation (Forage) Jan 2026

- Performed **exploratory data analysis (EDA)** and **predictive churn modeling** using **Python** and **SQL-driven analytical workflows**.

Deloitte Australia – Data Analytics Job Simulation (Forage) Oct 2025

- Executed data cleaning, **data visualization**, and business reporting on complex datasets for **strategic decision support**.

Education

VIT University 2023 – 2027
Bachelor of Technology in Computer Science & Engineering CGPA: 8.4

Certifications

AWS Certified Machine Learning Engineer – Associate | AWS Certified Data Engineer – Associate | Oracle Cloud Infrastructure 2025 Certified Data Science Professional| AI & Data Science Professional – IIT Roorkee (iHUB)
DevOps and Cloud Foundations – IBM | Introduction to Data Science – CISOC